

Trigger point and Myofascial Pain: The Bigger Picture and Tools for Care

Victoria P Williams, DO
Fall 2025



Objectives:

- Discuss the relationships between Trigger Points, Myofascial Pain Syndrome and Chronic Pain Syndrome
- Other Diagnostic consideration in Trp
- Discuss Mechanical Factors that can perpetuate TrP
- Self-Care and Prevention of Trp and Myofascial Pain Syndrome

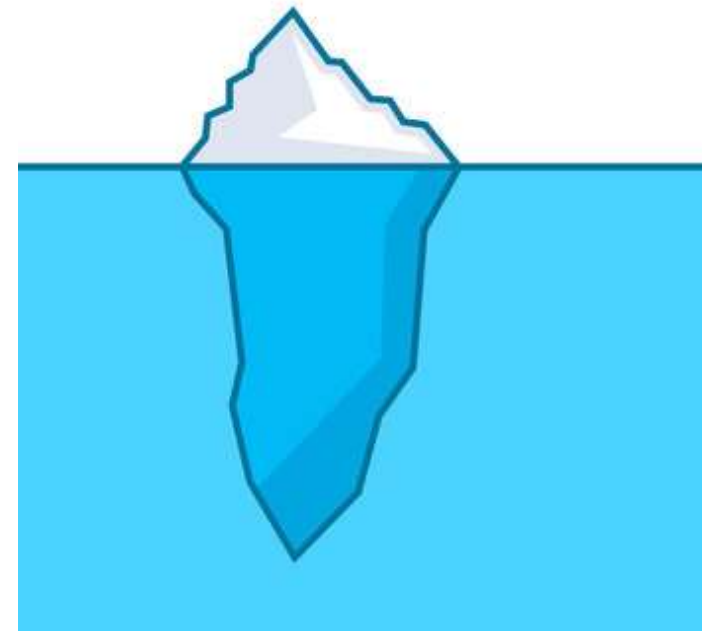


Trigger Points can be a symptom of a Bigger Picture

Trigger point: the local lesion (hyperirritable spot in a taut band with endplate dysfunction).

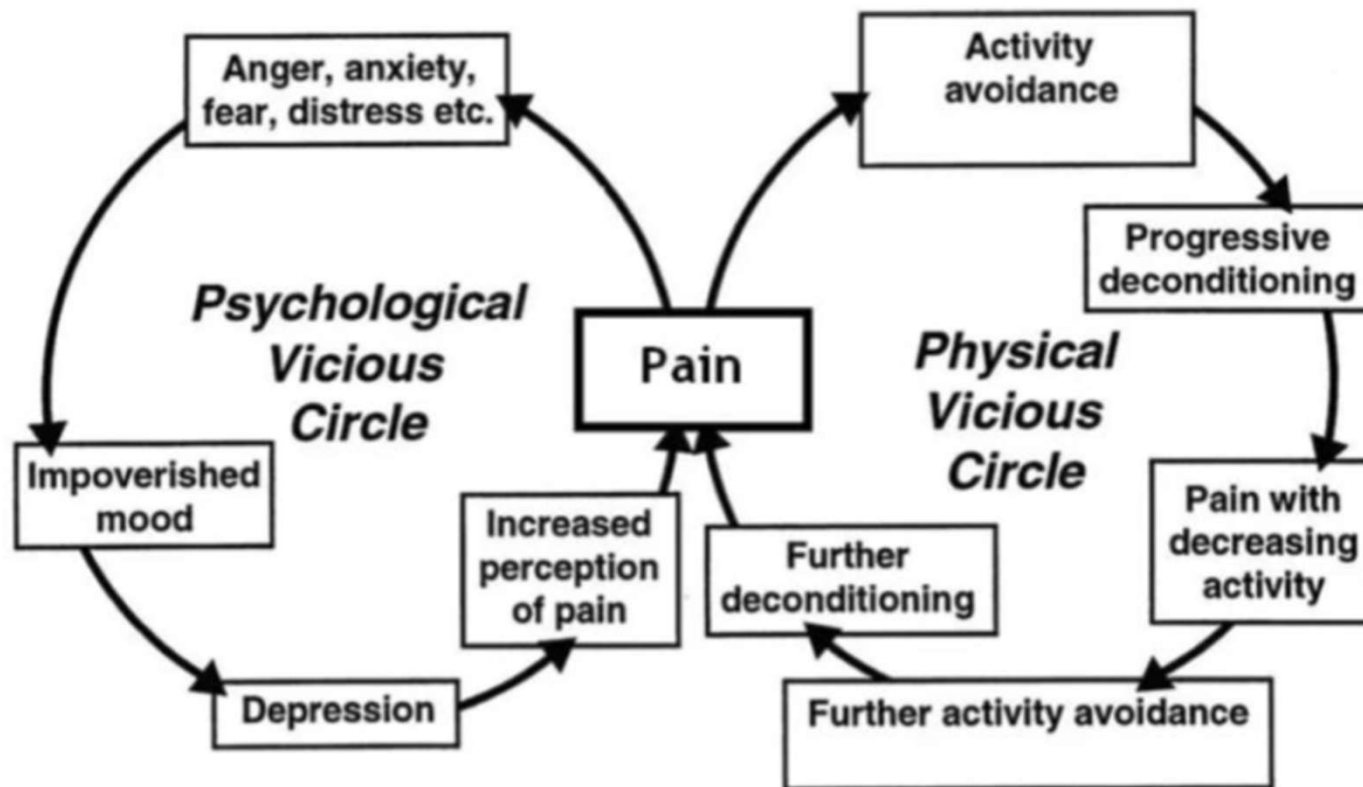
Myofascial Pain Syndrome: the broader syndrome that develops when trigger points drive persistent nociception, referred pain, motor dysfunction, and autonomic changes.

Chronic Pain Syndrome: A condition where pain persists beyond normal tissue healing time, driven more by changes in the nervous system and psychosocial factors than by the original injury. Persistent pain >3–6 months, maintained by central sensitization and systemic factors rather than the original lesion.



NS is sensitized

Trigger Points can be a symptom of a Bigger Picture



TRPT start from postural
in pain = don't want to move

Other Diagnostic Considerations

Hormonal Factors

- **Gonadal Hormones:** Estrogen can influence pain, while testosterone appears to have a protective, anti-pain effect.
- **Subclinical Hypothyroidism:** Even if thyroid levels are technically normal, this condition is associated with muscle pain and TrPs.

Sleep Deprivation:

- A lack of sufficient sleep is strongly associated with muscle pain and can be a perpetuating factor for myofascial pain syndrome.

Other Diagnostic Considerations

nutritional deficiencies create vulnerability

Vitamin D: Vitamin D is a prohormone essential for calcium and phosphorus regulation in bone.

- Vitamin D receptors are present in muscle tissue, providing a direct link to muscle function.
- Deficiency can lead to Type II, fast-twitch muscle fiber atrophy.
- A link has been shown between Vitamin D deficiency and chronic widespread pain, including fibromyalgia.

Magnesium:

- Magnesium may play a role in helping muscles relax, but its direct effect on TrPs in humans has not been demonstrated.

Iron levels:

- Iron deficiency is presented as a factor that can contribute to chronic myofascial pain, particularly in women.

At-Home Trigger Point Self-Care

goal is prevention not just chasing symptoms

Tools: A foam roller, lacrosse ball, or a massage cane can be used to apply direct pressure to the trigger point.

Locate: Find the taut band and the most tender spot within it. This is your trigger point.

Apply Pressure: Place the tool directly on the trigger point and apply sustained, firm pressure for 30-90 seconds. The goal is to feel the sensation "melt" or subside. Do not apply pressure that is unbearable or causes pain to radiate significantly.

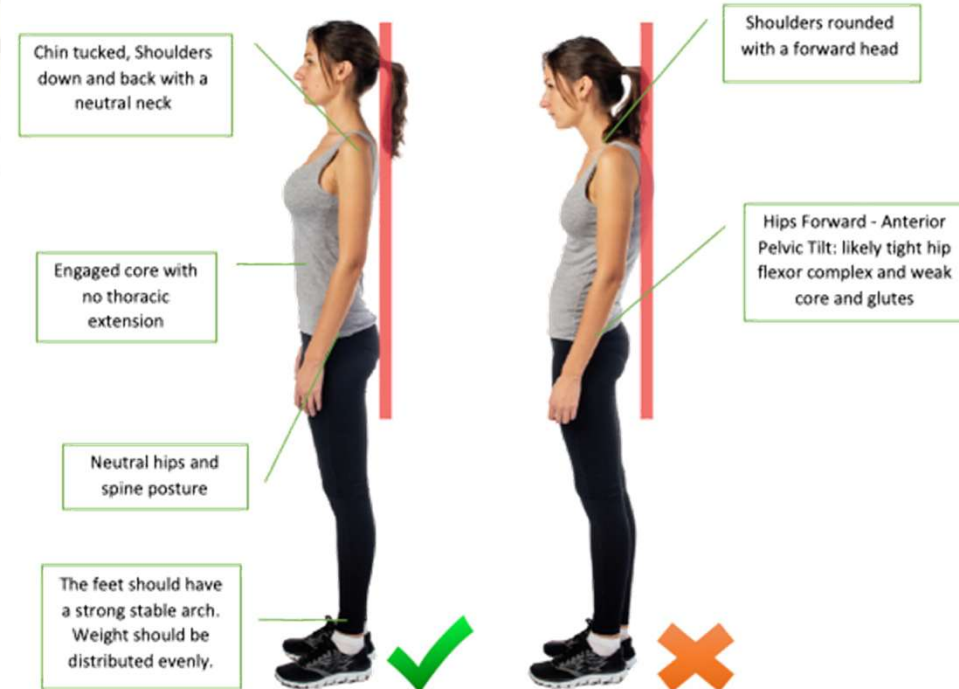
Stretch: After applying pressure, gently stretch the muscle to its full length.



Mechanical Perpetuating Factors:

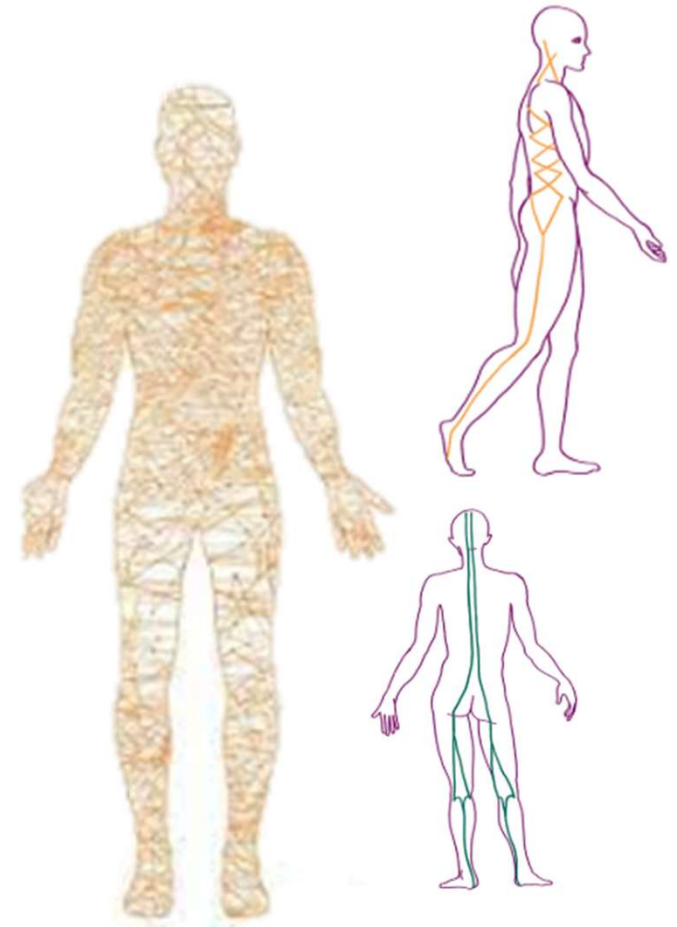
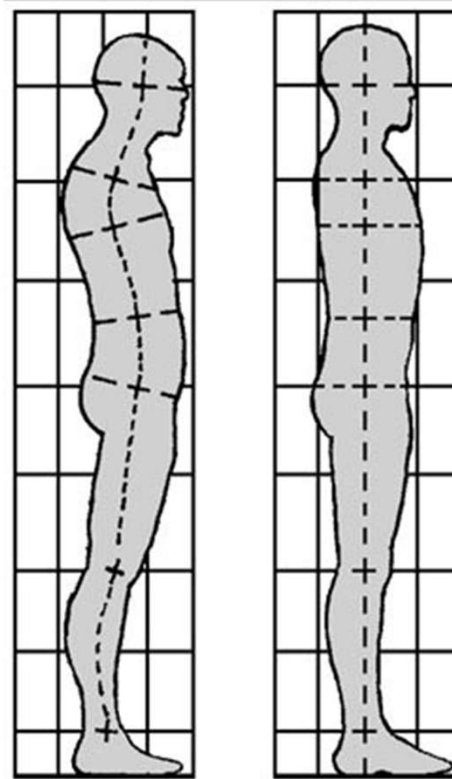
- Structural compensation
- Postural imbalance
- Repetitive use

Structural stresses occur because of muscle overuse to compensate for muscle imbalances caused by body structure asymmetry.



The role of FASCIA in Trigger Points

- Force transmission
- Pain force
- Density and hydration



pain may come from sensitized nerve in fascia - Travell

Forward Head Posture

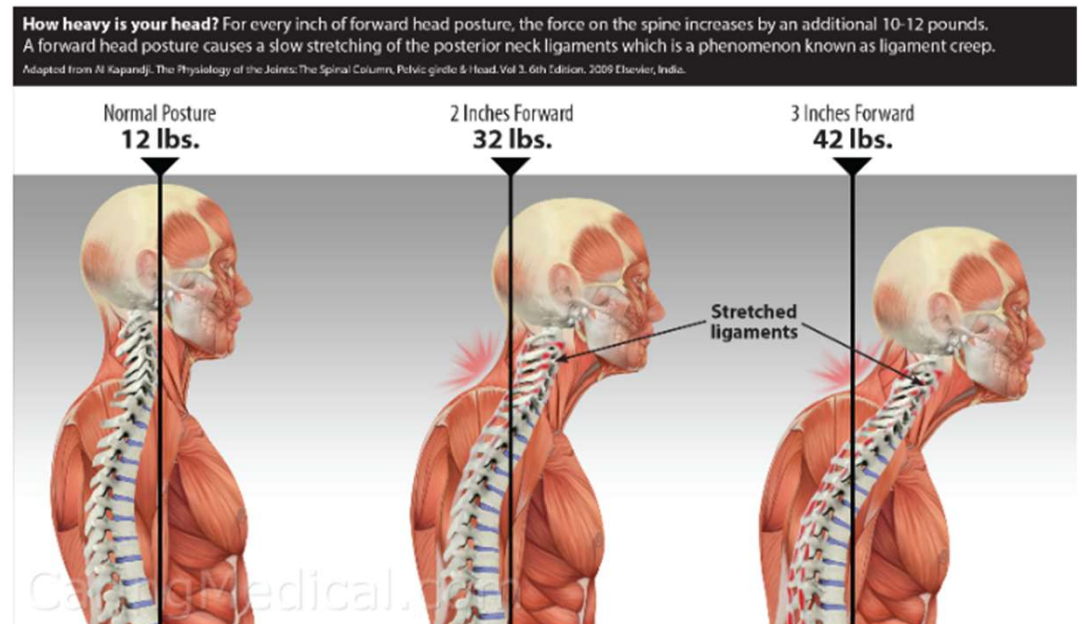
Forward Head Posture (FHP) forces neck muscles to work overtime, leading to muscle overload and trigger points.

This stress causes a chain reaction:

- Neck muscles in the back get short and tight.
- Muscles in the front become weak.
- This stresses the neck's joints.

FHP is a major risk factor for tension headaches and is strongly correlated with trigger points.

fascia transmits up to 40% of muscle force
creep - ligaments gradually lengthen leading to instability
circulation
can cause entrapment of ulnar nerve and radiate down arm

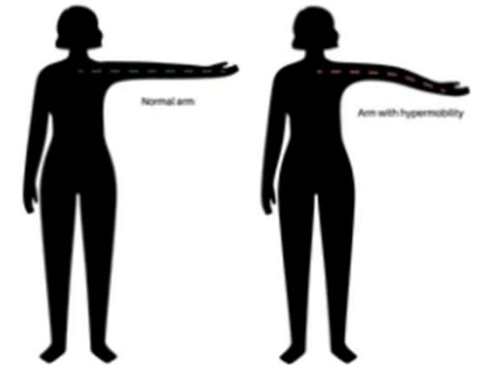


Adapted from Al Kapandji, *The Physiology of the Joints: The Spinal Column, Pelvic girdle & Head*. Vol. 3. 6th Edition. 2009 Elsevier, India

Hypermobility

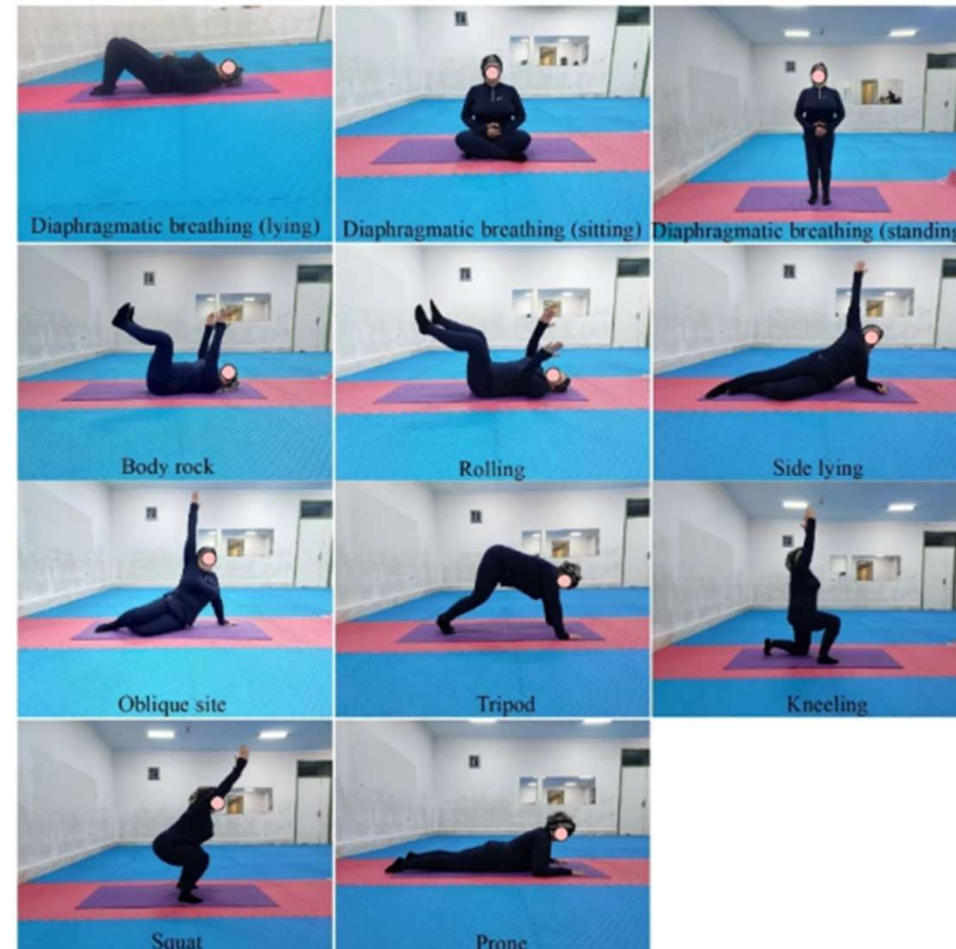
- Hypermobility:** Some people have increased collagen elasticity, leading to hypermobility.
- Mechanical Stress:** This hypermobility creates chronic instability, forcing muscles to work overtime to stabilize joints.
- TrP Risk:** The constant muscle strain makes these individuals more susceptible to trigger points and associated pain.

JOINT HYPERMOBILITY SYNDROME



Prevention and Empowerment

- Move Regularly
- Keep Your Core Strengthened
- Maintain Physiologic ROM
- Consider early treatment for TrPs

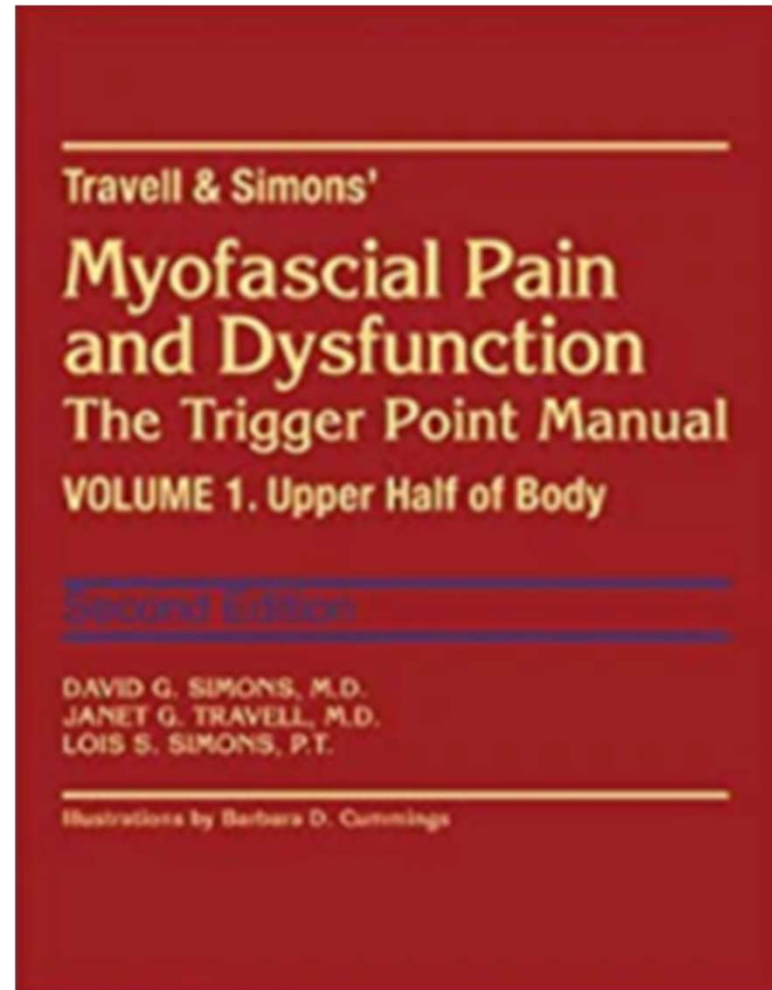


Prevention and Empowerment

- Hydrate and Optimize Fascia
- Address postural asymmetries early
- Nutrition, sleep, and hormonal balance
- Maintain a healthy narrative of pain
- Prevent CNS Hypersensitivity



References



take home
trigger points: local lesions driving
nociception
myofascial pain syndrome: broader syndrome
with referred pain and autonomic changes
chronic pain syndrome: pain amplified by
central sensitization and psychosocial factors
perpetuating factors: includes hormones,
sleep, nutrition, core strength, self-care tools
prevention and empowerment: movement,
hydration, nutrition, core strength, self-care
tools
treatment works best when both the local
lesion and systemic factors are addressed